

Songyou Peng | Curriculum Vitae

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Education

ETH Zurich <i>Doctor of Sciences, Max Planck ETH Center for Learning Systems PhD Fellowship</i> Supervisor: Prof. Marc Pollefeys & Prof. Andreas Geiger Committee: Prof. Leonidas J. Guibas (Stanford) and Prof. Vincent Sitzmann (MIT)	Zurich, Switzerland 09/2019–11/2023
Heriot-Watt University/University of Girona/University of Bourgogne <i>Erasmus Mundus M.Sc in Computer Visions and Robotics (VIBOT)</i> GPA: 17/20 (rank 3/23) with distinction Supervisor: Prof. Daniel Cremers	09/2015–09/2017
Xi'an Jiaotong University <i>B.Eng in Automation, focus: artificial intelligence</i>	Xi'an, China 08/2011–07/2015

Experience

Google DeepMind <i>Senior Research Scientist</i> - Co-lead of Vision Banana, a unified model that achieves SoTA on many vision tasks. - Core contributor of Nano Banana on pretraining and post training. - Core contributor of Gemini 3 & 2.5, and Project Astra, pre-/post-training for multi-modal spatial reasoning. - Core research, applications and products on 3D/4D reconstruction inside Google and Waymo.	San Francisco, USA 05/2024–present
ETH Zurich <i>Senior Researcher/Postdoc</i> - Advised 3 PhD students and 4 master students on their research projects. - Drafted, applied, and successfully obtained research fundings for 2 PhD positions.	Zurich, Switzerland 12/2023–05/2024
Google Research <i>Research Intern, mentor: Prof. Thomas Funkhouser</i> - Published OpenScene at CVPR 2023, first effort in open-vocabulary 3D scene understanding. - Directly resulted in a world-scale scene understanding effort inside Google called Geo Foundational Features.	Mountain View, USA 07/2022–11/2022
Meta Reality Labs Research <i>Research Intern, mentor: Dr. Michael Zollhöfer</i> Real-time neural rendering for 360-degree indoor scenes.	Pittsburgh, USA (remote) 09/2021–12/2021
Agency for Science, Technology and Research (A*STAR) <i>Research Engineer, Institute for Infocomm Research</i> - Performed an independent research project on universal architecture for bad-weather image restoration. - Worked on traffic flow prediction with gated spatial-temporal CNNs and graph CNNs.	Singapore 10/2018–07/2019
Advanced Digital Sciences Center, UIUC <i>Research Engineer, supervisor: Dr. Stefan Winkler, IEEE Fellow</i> Research in affective computing. - Developed a facial emotion analysis SDK for a 2-million SGD project. - Published an ACM MM demo paper and an IEEE Transactions on Affective Computing paper. - Won 1st place in vision-only task and 2nd place in overall in OMG-Emotion Challenge 2018.	Singapore 01/2018–03/2019
Technical University of Munich (TUM) <i>Master Thesis, supervisor: Prof. Daniel Cremers & Dr. Yvain Queau</i>	Munich, Germany 01/2017–07/2017

Depth Super-Resolution using photometric techniques.

- Proposed three photometric methods to obtain high-resolution depths with fine geometric details.
- One TPAMI paper and one ICCVW paper.

INRIA

Research Intern, supervisor: Prof. Peter Sturm

Grenoble, France

2016 & 2017 summer

- ICCV oral paper: designed a calibration guidance system for obtaining optimal calibration images.

Selected Publications (Full List at Google Scholar)

- **Project Co-Lead**, "Image Generators are Generalist Vision Learners", *arXiv*, 2026
- Youming Deng, **Songyou Peng**, Junyi Zhang, Kathryn Heal, Tiancheng Sun, John Flynn, Steve Marschner, Lucy Chai, "Selfi: Self Improving Reconstruction Engine via 3D Geometric Feature Alignment", *CVPR*, 2026
- Xianzheng Ma, TAO SUN, Shuai Chen, Yash Sanjay Bhalgat, Jindong Gu, Angel X Chang, Iro Armeni, Iro Laina, **Songyou Peng***, Victor Adrian Prisacariu*, "Do 3D Large Language Models Really Understand 3D Spatial Relationships?", *ICLR*, 2026
- **Songyou Peng** (core contributor) "Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multi-modality, Long Context, and Next Generation Agentic Capabilities", 2025
- **Songyou Peng** (core contributor) "Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multi-modality, Long Context, and Next Generation Agentic Capabilities", 2025
- Botao Ye, Sifei Liu, Haofei Xu, Xueting Li, Marc Pollefeys, Ming-Hsuan Yang, **Songyou Peng**, "No Pose, No Problem: Surprisingly Simple 3D Gaussian Splats from Sparse Unposed Images", *ICLR*, 2025. (**Oral, top 1.6%**)
- Jan Ackermann, Jonas Kulhanek, Shengqu Cai, Haofei Xu, Marc Pollefeys, Gordon Wetzstein, Leonidas Guibas, **Songyou Peng**, "CL-Splats: Continual Learning of Gaussian Splatting with Local Optimization", *ICCV*, 2025.
- Boyang Deng, **Songyou Peng**, Kyle Genova, Gordon Wetzstein, Noah Snavely, Leonidas Guibas, Thomas Funkhouser, "Visual Chronicles: Using Multimodal LLMs to Analyze Massive Collections of Images", *ICCV*, 2025. (**Highlight**)
- Jonas Kulhanek, **Songyou Peng**, Zuzana Kukelova, Marc Pollefeys, Torsten Sattler, "WildGaussians: 3D Gaussian Splatting in the Wild", *NeurIPS*, 2024.
- Rui Huang, **Songyou Peng**, Ayça Takmaz, Federico Tombari, Marc Pollefeys, Shiji Song, Gao Huang, Francis Engelmann, "Segment3D: Learning Fine-Grained Class-Agnostic 3D Segmentation without Manual Labels", *ECCV*, 2024.
- Weining Ren*, Zihan Zhu*, Boyang Sun, Jiaqi Chen, Marc Pollefeys, **Songyou Peng**, "NeRF *On-the-go*: Exploiting Uncertainty for Distractor-free NeRFs in the Wild", *CVPR*, 2024.
- **Songyou Peng***, Zihan Zhu*, Viktor Larsson, Zhaopeng Cui, Martin R. Oswald, Andreas Geiger, Marc Pollefeys, "NICER-SLAM: Neural Implicit Scene Encoding for RGB SLAM", *3DV*, 2024. (**Oral, Best Paper Honorable Mention**)
- **Songyou Peng**, Kyle Genova, Chiyu "Max" Jiang, Andrea Tagliasacchi, Marc Pollefeys, Thomas Funkhouser, "OpenScene: 3D Scene Understanding with Open Vocabularies", *CVPR*, 2023.
- **Songyou Peng***, Zihan Zhu*, Viktor Larsson, Weiwei Xu, Hujun Bao, Zhaopeng Cui, Martin R. Oswald, Marc Pollefeys, "NICE-SLAM: Neural Implicit Scalable Encoding for SLAM", *CVPR*, 2022.
- **Songyou Peng**, Chiyu "Max" Jiang, Yiyi Liao, Michael Niemeyer, Marc Pollefeys, Andreas Geiger, "Shape As Points: A Differentiable Poisson Solver", *NeurIPS*, 2021. (**Oral, top 0.6%**)

- **Songyou Peng**, Michael Niemeyer, Lars Mescheder, Marc Pollefeys, Andreas Geiger, “Convolutional Occupancy Networks”. *ECCV*, 2020. (**Spotlight, top 5%**)
- **Songyou Peng**, Peter Sturm, “Calibration Wizard: A Guidance System for Camera Calibration Based on Modelling Geometric and Corner Uncertainty”. *ICCV*, 2019. (**Oral, top 4.6%**)
- **Songyou Peng***, Bjoern Haefner*, Alok Verma*, Yvain Quéau, Daniel Cremers, “Photometric Depth Super-Resolution”. *TPAMI*, 2019.
- Zehao Yu, **Songyou Peng**, Michael Niemeyer, Torsten Sattler, Andreas Geiger, “MonoSDF: Exploring Monocular Geometric Cues for Neural Implicit Surface Reconstruction”, *NeurIPS*, 2022.
- Michael Oechsle, **Songyou Peng**, Andreas Geiger, “UNISURF: Unifying Neural Implicit Surfaces and Radiance Fields for Multi-View Reconstruction”. *ICCV*, 2021. (**Oral, top 3%**)
- Christian Reiser, **Songyou Peng**, Yiyi Liao, Andreas Geiger, “KiloNeRF: Speeding up Neural Radiance Fields with Thousands of Tiny MLPs”, *ICCV*, 2021.
- Shaohui Liu, Yinda Zhang, **Songyou Peng**, Boxin Shi, Marc Pollefeys, Zhaopeng Cui, “DIST: Rendering Deep Implicit Signed Distance Function with Differentiable Sphere Tracing”. *CVPR*, 2020.

Awards & Fellowships

- ECVA PhD Award (two awardees across the whole Europe per year) 2024
- Best Paper Honorable Mention Award at 3DV 2024
- Max Planck ETH Center for Learning Systems PhD Fellowship 2019 – 2023
- Best Presentation Award at ICVSS 2023
- 1st place in partial object recovery in SHARP Challenge at CVPR 2022
- Outstanding Reviewer of CVPR (Top 2%) 2022
- Highlighted Reviewer of ICLR (Top 8%) 2022
- Most Influential ECCV Papers: ConvONet #12 (link) 2020
- 1st place in vision-only task and 2nd in overall in OMG-Emotion Recognition Challenge 2018
- EU Erasmus+ mobility grant, awarded by European Union Commission 2016 & 2017
- Excellent bachelor thesis (top 5% of all graduates), XJTU 2015
- 1st in Search and Rescue Robot Challenge, California State University, USA 2010
- 2nd in Trinity College Fire Fighting Home Robot Contest, Connecticut, USA 2010
- 2nd in RoboCup Junior China Qualification Trial, Suzhou, China 2007

Invited Talks

- My 10-Year Journey in 3D Vision. *3DV PhD Award Talk* 2026
- Building Visual Intelligence. *Amazon FAR* 2025
- Building Visual Intelligence. *Meta* 2025
- A “Splatacular” Year of 3D Reconstruction. *Stanford University, KAIST* 2025
- 2D Magic in a 3D World. *Czech Technical University (CTU)* 2024
- 2D Magic in a 3D World. *Imperial College London* 2024
- 2D Magic in a 3D World. *The University of Hong Kong* 2024
- Dive into Neural Implicit-Explicit 3D Representations. *Invited lecture at SGP graduate school* 2023
- OpenScene: 3D Scene Understanding with Open Vocabularies. *Apple* 2023
- OpenScene: 3D Scene Understanding with Open Vocabularies. *Stability.ai* 2023

- OpenScene: 3D Scene Understanding with Open Vocabularies. *Peking University* 2023
- Learning to Reconstruct and Understand the 3D World. *Microsoft Mix Reality & AI Lab* 2023
- Learning Neural Scene Representations for 3D Reconstruction and Understanding. *Shanghai AI Lab* 2023
- How do NeRF and CLIP advance 3D Scene Reconstruction and Understanding? *Bosch* 2023
- Large-Scale 3D Scene Reconstruction with NeRF. *Stanford University* 2022
- Towards Practical Applications of NeRF. *Adobe Research* 2022
- Neural Scene Representations for 3D Reconstruction. *University of Basel* 2022
- Shape As Points: A Differentiable Poisson Solver. *Talking Papers Podcast* 2022
- Towards Practical Applications of NeRF. *GAMES Webinar Series* 2021

Teaching

Teaching Assistant at ETH Zurich

- [252-0579-00L] 3D Vision (Lecturer: Marc Pollefeys & Daniel Barath) Spring 23
- [263-5902-00L] Computer Vision (Lecturer: Marc Pollefeys & Siyu Tang & Fisher Yu) Fall 22
- [252-0579-00L] 3D Vision (Lecturer: Marc Pollefeys & Daniel Barath) Spring 22
- [263-5904-00L] Deep Learning for Computer Vision: Seminal Work Spring 22
- [252-0579-00L] 3D Vision (Lecturer: Marc Pollefeys & Viktor Larsson) Spring 20
- [263-5904-00L] Deep Learning for Computer Vision: Seminal Work Spring 20

Teaching Assistant at University of Tübingen

- [ML-4103] Deep Learning (Lecturer: Andreas Geiger) Winter 20/21

PhD Students Mentored at Google DeepMind

- Gene Chou (Cornell): CityRAG [arXiv'26] 2025-2026
- Youming Deng (Cornell): Selfi [CVPR'26 Oral] 2025
- Jiahao Wang (John Hopkins): Waymo world simulator [CVPR'26] 2025
- Jonas Kulhanek (CTU/ETH Zurich): LODGE [NeurIPS'25 spotlight paper] 2025
- Boyang Deng (Stanford): Visual Chronicles [ICCV'25 highlight paper] 2024
- Anh Thai (Georgia Tech): SplatTalk [ICCV'25 paper] 2024

Supervised Master Students at ETH Zurich

- [Semester project] Jan Ackermann (Next: PhD Student at Stanford University) 2024
- [Master thesis] Gonca Yilmaz (Next: Software Engineer at Google Zurich) 2024
- [Master thesis] Weining Ren (Next: PhD Student at the University of Hong Kong) 2023
- [Master thesis] Lei Li (Next: Research Engineer at ByteDance) 2023
- [Master thesis] Mirlan Karimov (Next: PhD Student at Mercedes-Benz AG) 2023
- [Semester project] Gonca Yilmaz (Next: Master thesis with CVG, ETH Zurich) 2023
- [Semester project] Shengqu Cai (Next: PhD Student at Stanford University) 2023
- [Semester project] Zihan Zhu (Next: PhD Student at ETH Zurich) 2022
- [Master thesis] Pfister Severin (Next: Consultant at McKinsey) 2021
- [Semester project] Weirong Chen (Next: PhD Student at TU Munich) 2021

PhD Committee Served

- Chamin Hewa Koneputugodage (Australian National University) 2025

Service

- **Area Chair:** NeurIPS 2026, ECCV 2026, CVPR 2026, ICLR 2026, ICCV 2025, ICML 2025, 3DV 2024 (done during PhD)
- **Publicity Chair:** 3DV 2025
- **Workshop Organizer:**
 - MUSI: 2nd Workshop on Multi-Modal Spatial Intelligence, CVPR 2026
 - MUSI: 1st Workshop on Multi-Modal Spatial Intelligence, ICCV 2025
 - 5th Workshop on 3D Scene Understanding for Vision, Graphics, and Robotics, CVPR 2025
 - OpenSUN3D: 3rd Open-Vocabulary 3D Scene Understanding, ECCV 2024
 - FOCUS: Foundation Models Creators Meet Users, ECCV 2024
 - OpenSUN3D: 2nd Open-Vocabulary 3D Scene Understanding, CVPR 2024
 - OpenSUN3D: 1st Open-Vocabulary 3D Scene Understanding, ICCV 2023
- **Conference Reviewer:** CVPR, ICCV, ECCV, SIGGRAPH, SIGGRAPH Asia, NeurIPS, ICLR, RSS
- **Journal Reviewer:** TPAMI, IJCV, CVIU